

LORM — Liability Object Representation Module

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Canonical Definition

The Liability Object Representation Module (LORM) governs how a Distinguished Liability Object is formally represented, described, structured, referenced, classified, and made consistently interpretable so that the same governed matter can be used across subsequent Liability governance without semantic drift, hidden assumptions, premature Liability conclusions, or loss of distinction from adjacent governed concepts.

LORM governs the transition:

Distinguished Liability Object → Representation Formation → Semantic Structuring → Representation Validation → Represented Liability Object
Its foundational distinction is:

Representation of the Liability Object ≠ Liability Determination about the Liability Object.

Operational Role

The operational role of LORM is:

to convert a sufficiently identified, bounded, and distinguished Liability Object into a stable governed representation that can be consistently understood and referenced throughout the remaining LIGA™ lifecycle.

LOIM identifies the Object Candidate.

LOBM establishes its boundary.

LODM establishes its distinction.

LORM establishes its formal representation.

Its central operational question is:

How must this Liability Object be represented so that every later governance stage refers to the same governed matter without embedding an ungoverned Liability conclusion?

Position in the Parent Standard Flow

The internal LOS progression is:

Object Identification → Object Boundary → Object Distinction → Object Representation → Object Continuity
LORM governs:

Object Representation

Its principal input is:

Distinguished Liability Object

Its principal output is:

Represented Liability Object

The canonical transition is:

Distinguished Liability Object → Object Representation Governance → Represented Liability Object → Object

Continuity Governance

Governed Space

LORM governs Liability Object Representation.

Its governed space includes:

Object Naming, Object Description, Object Identity Reference, Object Type, Object Classification, Object Boundary Representation, Object Component Representation, Object State, Object Context Association, Object Distinction References, Object Uncertainty Representation, Object Assumption Representation, Composite Object Representation, Multi-Object Representation, representation granularity, semantic neutrality, machine interpretability, representation consistency, representation correction, and representation readiness.

LORM represents the Object.

It does not determine Liability.

Representation Formation

A Distinguished Liability Object must be transformed into an explicit governed representation.

The representation should make the Object understandable without requiring later governance stages to reconstruct its meaning independently.

The canonical progression is:

Distinguished Liability Object

↓

Representation Elements Identified

↓

Canonical Object Description Formed

↓

Boundary & Distinction References Attached

↓

State / Uncertainty / Assumptions Represented

↓

Representation Consistency Tested

↓

Represented Liability Object

Representation Purpose

The purpose of Object Representation is not merely documentation.

It provides a stable semantic reference for later LIGA™ governance.

A sufficiently governed representation should allow later stages to answer:

Which Object is being evaluated?

What constitutes it?

What does not constitute it?

How is it identified?

What type of matter is it?

Which Context does it belong to?

Which uncertainty remains?

Which assumptions affect its current representation?

Without this stability, later Liability governance may unintentionally operate on different interpretations of the same Object.

Canonical Object Representation

A Canonical Object Representation should provide a single authoritative governed representation of the Liability Object for the relevant governance state.

It may contain:

Object Identity

Canonical Object Name or Reference

Canonical Object Description

Object Type

Object Classification

Object Boundary Reference

Material Components

Relevant Context Reference

Object Distinction References

Object State

Uncertainty State

Material Assumptions

Representation Version

where applicable.

The canonical representation does not eliminate underlying evidence or source records.

It provides the governed methodological representation of the Object.

Object Naming

A Liability Object may require a human-readable name.

The name should:

identify the matter clearly, remain sufficiently neutral, avoid ungoverned conclusions, and remain distinguishable from adjacent Objects.

Names such as:

“negligent maintenance failure,” “illegal system action,” or “liable product defect” may improperly encode conclusions not yet established.

Therefore:

Object Name ≠ Liability Conclusion.

Object Description

The Object Description should explain what the governed matter is.

It should be:

specific, bounded, neutral, traceable, and consistent with prior LOS governance.

The description must not silently expand or contract the Object Boundary.

Therefore:

Object Description ≠ Authority to Redefine the Object.

Where the description reveals a boundary problem, the matter should return to LOBM rather than changing the Object silently through wording.

Object Identity Reference

The representation must maintain a stable reference to Object Identity.

The reference may remain constant even where:

the description changes, the representation is corrected, new information emerges, or the Object state changes.

Therefore:

Object Identity ≠ Object Representation Version.

One Object may have multiple representation states while maintaining one continuous Object Identity.

Object Type

LORM may represent an Object Type to support interpretation and governance.

Possible types may include:

action, omission, transaction, decision, product-related matter, service-related matter, system operation, system state, contract-related matter, condition, process, sequence, composite matter, or another appropriate type.

Object Type describes the nature of the governed matter.

It does not determine Liability.

Therefore:

Object Type ≠ Liability Type.

Object Classification

An Object may require one or more methodological classifications.

Classification may assist:

routing, comparison, interoperability, analytics, automation, or later governance.

However:

Object Classification ≠ Liability Classification.

Classification must not be used to infer automatically:

Liability Basis, Liability Attachment, Liability Scope, or Liability Allocation.

Object Boundary Representation

The Canonical Object Representation should reference the established Object Boundary.

Where useful, it may represent:

included elements, excluded but related elements, open boundaries, boundary uncertainty, shared components, cross-environment extent, and temporal extent.

The representation must not silently alter those states.

Therefore:

Boundary Representation $\neq$ Boundary Governance.

LOBM owns the Object Boundary.

LORM represents its governed output.

Component Representation

Composite Liability Objects may contain multiple components.

LORM must permit explicit representation of:

component identities, component roles, component relationships, shared components, and component status.

A component must not be mistaken for the complete Object.

Therefore:

Object Component $\neq$ Complete Liability Object by default.

Composite Object Representation

A Composite Liability Object should be represented in a way that preserves both:

the identity of the composite whole, and the identities of materially relevant components.

The canonical structure may be:

Composite Object

→ Component A → Component B → Component C The representation should make clear why the components belong to one Object without treating their aggregation as proof of Liability.

Multiple Object Representation

One Liability Context may contain multiple Liability Objects.

LORM must ensure that each Object remains separately representable.

The representation should prevent:

Object A attributes being assigned to Object B, Object B evidence being treated automatically as Object A evidence, or shared context being interpreted as shared Object Identity.

Therefore:

Shared Context $\neq$ Shared Object Representation.

Object State Representation

LORM should represent the current Object Governance State.

Possible states may include:

Identified

Bounded

Distinguished

Represented

Uncertain

Disputed

Under Reassessment

Superseded

or another controlled state.

Object State describes the state of Object Governance.

It does not describe whether Liability exists.

Therefore:

Object State $\neq$ Liability State.

Context Association

Every represented Liability Object should remain traceably associated with its Governed Liability Context.

The representation should identify the relevant Context without collapsing Object and Context.

Therefore:

Context Association $\neq$ Context Identity.

The Object remains a distinct governed object within or in relation to its Context.

Distinction References

Where materially useful, the representation should preserve explicit distinction references showing that the Object is not identical to adjacent governed concepts.

Examples may include:

Object $\neq$ Harm Object $\neq$ Claim Object $\neq$ Liability Subject Object $\neq$ Operational Mechanism Object $\neq$ Liability Basis

These references can prevent downstream semantic drift.

Neutral Representation

The Liability Object must be represented neutrally.

The representation must not presume:

fault, breach, negligence, wrongdoing, causation, responsibility, accountability, Liability, or legal outcome unless such states have already been independently governed and are legitimately required as contextual references.

The canonical principle is:

Representation Must Not Predetermine Liability.

Descriptive Fact and Evaluative Conclusion

LORM must distinguish descriptive representation from evaluative conclusion.

For example:

“System executed transaction X at T1” is structurally different from:

“System wrongfully executed transaction X.” The second formulation contains an evaluative conclusion.

Therefore:

Description of Object $\neq$ Evaluation of Object.

This distinction is especially important before Liability Basis and Determination governance.

Assertion and Representation

A Claim, participant, authority, organization, or other source may assert a description of the Object.

That assertion may be relevant.

It does not automatically become the Canonical Object Representation.

Therefore:

Asserted Description $\neq$ Canonical Object Representation.

Where assertions are retained, they should remain attributable to their source.

Evidence and Representation

Evidence may support the Object Representation.

However:

Evidence $\neq$ Object Representation.

Evidence describes or supports aspects of the Object.

LORM structures the governed representation.

ADIT® governs evidence integrity, provenance, and auditability.

LORM should preserve references to relevant governed evidence where needed without absorbing evidence governance.

Representation and Operational Reality

ORA™ may establish the Operational Reality underlying the Object.

LORM may represent relevant ORA™ outputs.

However:

Operational Reality Representation $\neq$ Liability Object Representation by default.

The Liability Object is represented specifically for Liability governance while preserving the original governed meaning of external objects.

Representation and Harm

Where Harm is relevant, LORM may reference a governed HAGA™ Harm object or state.

It must not silently represent the Harm as the Liability Object unless LOS has explicitly established that relationship.

Therefore:

Harm Representation $\neq$ Liability Object Representation by default.

Representation and Claims

A Claim may contain its own representation of the matter.

LORM must preserve the distinction between:

Claim Representation

and:

Canonical Liability Object Representation.

A Claim may change without requiring the Liability Object Representation to change.

Likewise, the Object Representation may be corrected without altering the Claim itself.

Representation Across Systems

The same Liability Object may need to be represented across:

governance platforms, databases, AI systems, audit systems, claims systems, organizational systems, or external interfaces.

Interoperability requires preservation of Object Identity and meaning across those representations.

Therefore:

Representation Transfer $\neq$ Object Transformation.

A change in technical format must not silently change the governed meaning of the Object.

Human-Readable Representation

The Object should be understandable to authorized human users.

Human-readable representation should clearly communicate:

what the Object is, its relevant boundary, its principal components, its state, and material uncertainty.

Clarity must not be achieved by replacing methodological precision with simplified but misleading terminology.

Machine-Readable Representation

Where machine processing is used, the representation should support structured interpretation of:

Object Identity, Object Type, Object Boundary references, Context references, component relationships, Object State, uncertainty, assumptions, and representation version.

Machine readability must preserve the same governed meaning as the human-readable representation.

Therefore:

Machine-Readable Representation $\neq$ Separate Object Meaning.

Representation Consistency

Different representations of the same Liability Object must remain semantically consistent.

A database record, human-readable report, API object, AI-readable structure, or audit record may use different formats.

Those formats must not create different Object meanings.

Therefore:

Format Difference $\neq$ Meaning Difference.

Where representations conflict materially, reconciliation is required.

Representation Granularity

Representation should provide enough detail for downstream governance without reproducing the entire Liability Context.

Too little detail may create ambiguity.

Too much detail may obscure the Object by importing unrelated context.

Therefore:

More Representation Detail $\neq$ Better Representation by default.

The correct level is:

Sufficient Semantic Granularity.

Representation of Uncertainty

Where Object meaning, boundary, classification, or another representation element remains uncertain, the uncertainty must be explicit.

Possible states may include:

Established

Partially Established

Uncertain

Disputed

Inferred

Under Reassessment

The representation must not convert uncertainty into certainty merely for technical convenience.

Therefore:

Structured Representation $\neq$ Established Certainty.

Representation of Assumptions

Material assumptions affecting Object Representation must remain visible.

Examples may include:

assumed component identity, assumed sequence membership, assumed system state, or assumed relationship between object elements.

Therefore:

Represented Assumption $\neq$ Established Fact.

Assumptions should remain distinguishable from verified representation elements.

Representation Conflict

Two systems, participants, or records may contain materially different representations of the same Object.

LORM should identify such conflict.

The system should not resolve the conflict merely by selecting the latest or most convenient description.

Possible responses include:

reconciliation, validation, return to Object Distinction, return to Object Boundary, or preservation of competing representations.

Therefore:

Representation Conflict $\neq$ Automatic Canonical Replacement.

Representation Drift

Representation Drift occurs when the description or structured representation of an Object changes incrementally while the change is not explicitly governed.

This may alter:

scope, meaning, components, classification, or implied Liability interpretation.

LORM must detect and prevent silent Representation Drift.

Therefore:

Representation Change Must Be Governed.

Representation Correction

A representation may later be shown to be inaccurate.

LORM must permit correction.

The canonical pattern is:

Representation V1 $\rightarrow$ Error Identification $\rightarrow$ Corrected Representation V2 The correction should preserve:

the earlier representation, the reason for correction, the corrected element, and the relationship between versions.

Therefore:

Representation Correction $\neq$ Historical Erasure.

Representation Reassessment

New information may require reassessment of the representation.

Reassessment may:

change the description, change classification, change represented components, change uncertainty, or reveal that Boundary or Distinction governance must be reopened.

LORM must route such issues appropriately.

It must not solve a Boundary or Distinction problem merely by rewriting the representation.

Therefore:

Representation Reassessment $\neq$ Authority to Redefine the Object.

Representation Sufficiency

The Object Representation is sufficient when later governance can reliably determine:

which Object is being referenced, what it means, what constitutes it, what remains outside it, which Context it belongs to, which material uncertainty remains, and which representation state is current.

Perfect descriptive completeness is not required.

Therefore:

Representation Sufficiency $\neq$ Exhaustive Description.

Representation Readiness

A Represented Liability Object is ready for Object Continuity governance when:

Object Identity is stable, the Object is neutrally described, its Boundary is represented, its distinctions are preserved, its relevant components are represented, uncertainty and assumptions are visible, and representation conflicts are sufficiently controlled.

Readiness does not establish Liability.

Minimum Implementation Framework

A conforming implementation of LORM must provide at least the following eight capabilities:

Canonical Object Representation

Capability to create a stable governed representation containing Object Identity, description, type, classification, state, Context association, and other materially necessary representation elements.

Boundary & Component Representation

Capability to represent Object Boundary, included components, excluded-but-related elements, Composite Object structure, and multiple Object relationships without silently changing the governed boundary.

Semantic Neutrality & Distinction Preservation

Capability to represent the Object without embedding ungoverned conclusions and to preserve its distinction from Harm, Claims, Subjects, Basis, mechanisms, consequences, and other adjacent concepts.

Uncertainty & Assumption Representation

Capability to represent uncertainty, disputes, inference, assumptions, and incomplete states without converting them into false certainty.

Human & Machine Interpretability

Capability to provide semantically consistent human-readable and machine-readable representations suitable for interoperable governance use.

Cross-System Representation Consistency

Capability to preserve Object Identity and meaning across different systems, formats, interfaces, and technical representations and to detect material representation conflicts.

Representation Change, Correction & Reassessment

Capability to govern representation changes, corrections, conflicts, and reassessment while preserving prior states and routing Boundary or Distinction issues back to the appropriate governance function.

Representation Sufficiency & Readiness Determination

Capability to determine when the Object Representation is sufficiently stable, neutral, interpretable, and traceable for Object Continuity governance.

Governance Logic

The canonical LORM governance logic is:

Distinguished Liability Object

↓

Establish Canonical Object Identity Reference

↓

Create Neutral Object Name and Description

↓

Represent Object Type / Classification / State

↓

Represent Object Boundary

↓ Represent Components / Composite Structure where applicable ↓

Associate Governed Liability Context

↓

Preserve Critical Object Distinctions

↓

Represent Uncertainty and Assumptions

↓

Establish Human-Readable and Machine-Readable Forms

↓

Test Cross-System Semantic Consistency

↓

Detect Representation Conflict / Drift

↓

Assess Representation Sufficiency

If insufficient:

Clarify / Reconcile / Correct / Reassess / Return to Boundary or Distinction Governance

If sufficient:

↓

Represented Liability Object

↓

Handoff to Object Continuity Governance

Materially new information may trigger:

Representation Reassessment → Corrected or Revised Representation without silently redefining the Object.

Governance Outputs

LORM may produce:

Canonical Liability Object Representation

Object Identity Reference

Canonical Object Name

Canonical Object Description

Object Type

Object Classification

Object State Representation

Object Boundary Reference

Object Component Representation

Composite Object Representation

Context Association Record

Object Distinction Reference Set

Object Uncertainty State

Object Assumption Record

Human-Readable Object Representation

Machine-Readable Object Representation

Representation Conflict State

Representation Change Record

Representation Correction Record

Representation Version

Representation Sufficiency State

The principal output is:

Represented Liability Object

The canonical handoff is:

Represented Liability Object → Object Continuity Governance

Critical Distinctions

LORM preserves the following distinctions:

Representation of the Liability Object ≠ Liability Determination about the Liability Object
Object Name ≠ Liability Conclusion
Object Description ≠ Authority to Redefine the Object
Object Identity ≠ Object Representation
Version Object Type ≠ Liability Type
Object Classification ≠ Liability Classification
Boundary Representation ≠ Boundary Governance
Object Component ≠ Complete Liability Object by default
Shared Context ≠ Shared Object Representation
Object State ≠ Liability State
Context Association ≠ Context Identity
Description of Object ≠ Evaluation of Object
Asserted Description ≠ Canonical Object Representation
Evidence ≠ Object Representation
Operational Reality Representation ≠ Liability Object Representation by default
Harm Representation ≠ Liability Object Representation by default
Claim Representation ≠ Canonical Liability Object Representation
Transfer ≠ Object Transformation
Machine-Readable Representation ≠ Separate Object
Meaning Format Difference ≠ Meaning Difference
More Representation Detail ≠ Better Representation by default
Structured Representation ≠ Established Certainty
Represented Assumption ≠ Established Fact
Representation Conflict ≠ Automatic Canonical Replacement
Representation Correction ≠ Historical Erasure
Representation Reassessment ≠ Authority to Redefine the Object
Representation Sufficiency ≠ Exhaustive Description

Cross-Architecture Boundary

LORM may represent references to governed outputs from other OOF® architectures without absorbing their governed meaning.

LCS provides the Governed Liability Context.

ORA™ may provide Operational Reality objects and states relevant to the Object.

HAGA™ may provide governed Harm references.

AGA™ may provide Authority, Responsibility, Accountability, or Attribution references.

ADIT® governs evidence supporting or documenting the representation.

VALIDOS™ governs Validation of relevant inputs.

OBIDENTITY® may provide persistent Identity, Origin, Ownership, Provenance, Authority, Permission, Transfer, or Continuity references.

AIG® may provide AI behavior objects or states.

ASGA™ may provide autonomous-system objects or states.

SIMULOS® may provide Simulation objects or states.

PGA™ may provide Prediction objects.

VFM® may provide Value objects or states.

LORM may reference these objects.

It does not redefine them.

The governing principle is:

Representation for Liability Governance does not transfer ownership of the represented object from its original governed domain.

Canonical Boundary

LORM begins with a:

Distinguished Liability Object

and governs its formal representation for continued LIGA™ use.

LORM governs:

Object Naming, Object Description, Object Identity Reference, Object Type, Object Classification, Object Boundary Representation, Object Component Representation, Composite Object Representation, Object State Representation, Context Association, Distinction References, Uncertainty Representation, Assumption Representation, human readability, machine readability, cross-system consistency, Representation Conflict, Representation Drift, Representation Correction, Representation Reassessment, and Representation Sufficiency.

LORM does not establish or redefine:

Object Boundary, Object Distinction, Object Continuity, Liability Basis, Liability Subjects, Liability Relationships, Liability Conditions, Liability-Relevant Connection, Liability Attachment, Liability Determination, Liability Scope, Liability Allocation, Transfer, Recourse, Resolution, or Finality.

LORM ends when the Liability Object is sufficiently represented for Object Continuity governance.

Use Case 1 — AI-Generated Operational Action

An AI-enabled system executes an operational action that becomes the subject of Liability Object governance.

LOIM identifies the relevant Object Candidate.

LOBM establishes the Object Boundary.

LODM distinguishes the Object from:

the AI system, the resulting Harm, the wider incident, the Claim, and the potential Liability Subjects.

LORM then creates a neutral representation of the governed Object.

Instead of describing it as:

“the AI system's negligent action,” the representation records the specific execution sequence, relevant Object Boundary, Context reference, system state, uncertainty, and Object Identity without assuming negligence or Liability.

Later LIGA™ standards can therefore evaluate the same Object without inheriting a premature conclusion.

Use Case 2 — Industrial Equipment Matter

A Liability Object concerns a specific equipment state and maintenance-related sequence.

Several systems refer to the matter differently:

the maintenance system uses a work-order number, the operational system uses an incident identifier, the supplier uses a component reference, and a Claim describes the matter as a product defect.

LORM establishes one Canonical Liability Object Representation linking those references while preserving their distinct meanings.

The technical identifiers may differ.

The governed Object meaning does not.

This allows later Liability governance to operate on one consistently represented matter.

Canonical Closing Statement

A Liability Object that cannot be represented consistently cannot remain stable across a complex Liability lifecycle.

Different people, systems, organizations, Claims, evidence records, and AI processes may describe the same matter differently. Those representations may vary in format, terminology, granularity, certainty, and purpose.

LORM creates a governed semantic anchor.

It allows the Liability Object to be represented across systems and stages without allowing description to become determination, formatting to become meaning, or assertion to become fact.

A representation may describe the object. It must not decide the Liability attached to it.

Represent the same object consistently before asking different governance systems to reason about it.

Governed Links

Liability Object Standard (LOS)

LIGA™ Architecture Map

LOCM — Liability Object Continuity Module

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